DAY 3: DNS SERVER CONFIGURATION

DNS Zone Setup, Records, Verification, and Troubleshooting

Introduction: How DNS Works

DNS stands for Domain Name System. It translates a domain name, such as www.musicmania.in, into an IP address that a device can use to connect to a server. An A record maps a name to an IPv4 address. A CNAME record creates an alias that points one DNS name to another name.

Important lab note: musicmania.in is used here as a practice domain. A locally configured zone works only when the client queries this DNS server and the network is configured to reach it. It will not automatically become publicly available on the Internet.

Task 1: Install and Set Up a DNS Server and Create a Zone

Option used in this report: BIND on Linux. Commands may need administrator/root privileges.

Steps:

1. Install BIND using the package manager for your Linux distribution. On Ubuntu/Debian, a common command is: sudo apt update && sudo apt install bind9 bind9-utils dnsutils
2. Check the service status: sudo systemctl status bind9
3. Create a zone entry in the BIND local configuration, commonly /etc/bind/named.conf.local. Example:

```
zone "musicmania.in" {
    type master;
    file "/etc/bind/db.musicmania.in";
};
```

4. Create the zone file at /etc/bind/db.musicmania.in. A sample file is shown below; adjust the serial number when changing the zone.

```
$TTL 3600
@    IN  SOA ns1.musicmania.in. admin.musicmania.in. (
         2026101301 ; serial
         3600       ; refresh
         900        ; retry
         604800     ; expire
         3600       ; negative cache TTL
)
@        IN  NS      ns1.musicmania.in.
ns1      IN  A       192.168.1.2
www      IN  A       192.168.1.50
play     IN  CNAME   www.musicmania.in.
```

5. In this example, the DNS server is assumed to use 192.168.1.2. If your server has a different IP address, replace it in the sample and make sure the client can reach it.

6. Validate the configuration and zone file before restarting:

- `sudo named-checkconf`
- `sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in`

Task 2: Add an A Record and Verify with nslookup or dig

The zone file contains this A record:

```
www     IN     A     192.168.1.50
```

After saving the configuration, validate it and reload BIND:

- `sudo named-checkconf`
- `sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in`
- `sudo systemctl reload bind9`

Test directly against the DNS server (replace 192.168.1.2 if needed):

- `nslookup www.musicmania.in 192.168.1.2`
- `dig @192.168.1.2 www.musicmania.in A`

Expected result: the answer section should show www.musicmania.in resolving to 192.168.1.50. This is the expected lab result; record the output actually returned by your server.

Task 3: Add and Test a CNAME Record

Add this record to the musicmania.in zone file:

```
play     IN     CNAME     www.musicmania.in.
```

Then validate and reload the zone:

- `sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in`
- `sudo systemctl reload bind9`
- `dig @192.168.1.2 play.musicmania.in A`
- `nslookup play.musicmania.in 192.168.1.2`

Expected result: play.musicmania.in is an alias of www.musicmania.in, which has the A record 192.168.1.50. The DNS response may show the CNAME and the target A record, depending on the query and resolver. The CNAME itself points to a name, not directly to an IP address.

Task 4: DNS Troubleshooting Procedure

If musicmania.in does not resolve correctly, I would troubleshoot in this order:

1. Check the DNS service status

```
sudo systemctl status bind9
```

Confirm that the service is active. If it is stopped or failed, inspect the error details.

2. Check the configuration syntax

```
sudo named-checkconf
```

This checks BIND configuration files for syntax errors.

3. Check the zone file

```
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
```

This checks the zone file and its records. Fix errors before reloading.

4. Review service logs

```
sudo journalctl -u bind9 --since '10 minutes ago'
```

Look for zone-loading errors, permission issues, or port-binding problems.

5. Restart or reload the service

```
sudo systemctl restart bind9
```

Restart only after correcting configuration errors; then check the status again.

6. Query the server directly

```
dig @192.168.1.2 www.musicmania.in A  /  nslookup www.musicmania.in 192.168.1.2
```

Check whether the authoritative lab server returns the expected address.

7. Check client DNS settings and connectivity

```
ipconfig /all (Windows) or resolvectl status (Linux); ping 192.168.1.2 if ICMP
is allowed
```

Make sure the client is configured to use the correct DNS server and can reach it. A failed ping alone does not prove DNS is broken.

Additional checks: confirm that TCP and UDP port 53 are allowed by the local firewall, the zone filename matches the configuration, the DNS server IP is correct, and the zone serial number is increased after edits.

Conclusion

In this practical, I learned how to create a DNS zone, configure A and CNAME records, verify name resolution with nslookup or dig, and troubleshoot common DNS service and configuration problems. The example records show the expected lab setup; actual results and screenshots should be added after running the commands on my own DNS server.